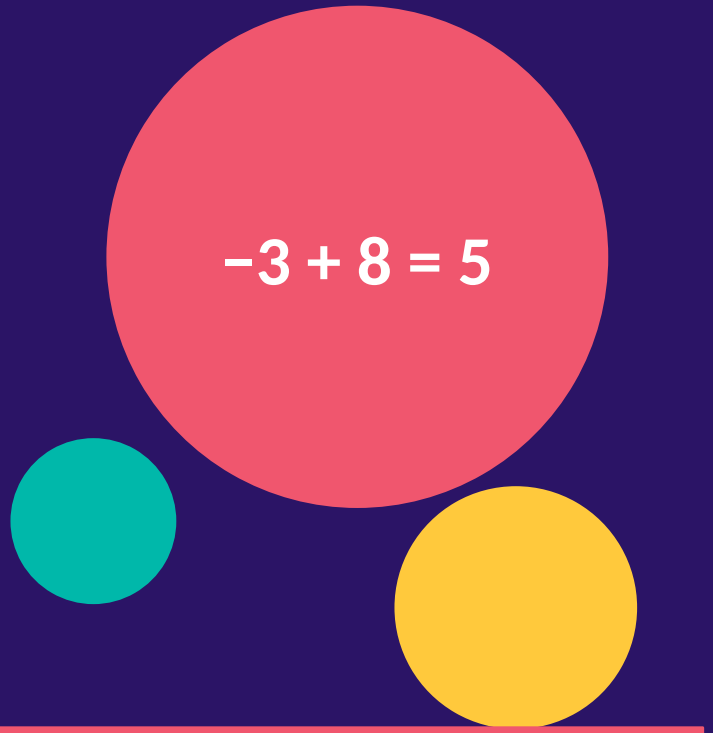
Negative numbers



Year 7 Set 3 · White Rose KS3 · 60 minutes · Mastery lesson

LO1: Add and subtract with negative numbers
LO2: Subtract a negative number
LO3: Multiply and divide with negative numbers

Lesson for Deniz · 60 minutes · OnMaths · Mr Ali

Have ready: mini whiteboard, pen, calculator OFF

Do Now

8 minutes

On your own. Show your working. Then we check together.

L3: Round 4.68 to 1 decimal place

L3: Round 0.072 to 1 significant figure

L3: Estimate 49 × 6.1

Skill: Which is colder, −4 °C or −9 °C?

Talk Task

3 minutes

Always, Sometimes or Never?
“Subtracting makes a number smaller.”

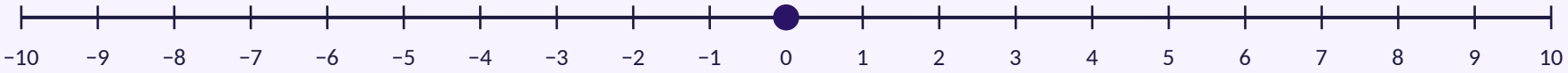
Say: I think it is _____ because _____ . For example, _____ .

Key facts: negative numbers

Key facts

← colder / smaller

warmer / bigger →



+ a number move RIGHT

− a number move LEFT

− (−3) same as + 3

× or ÷, same signs answer is positive

× or ÷, different signs answer is negative

LO1 I do: add and subtract

I do

Work out −3 + 8



LO1 I do: example 2

I do

Work out 4 − 9



You say it first, then click.

$-2 + 5$

$3 - 7$

$-6 - 5$

$-8 + 8$

LO1 Mini whiteboards (1 of 2)

MWB

Write your answer, show me, then click.

$-1 + 6$

$2 - 10$

LO1 Mini whiteboards (2 of 2)

MWB

Write your answer, show me, then click.

The temperature is −2 °C. It falls by 7 °C. What is the new temperature?

What is the difference between −4 °C and 6 °C?

LO1 You do

B · S · G · P

BRONZE

$-5 + 9$

SILVER

$7 - 12$

GOLD

$-3 - 4 + 10$

PLATINUM

6 am: −6 °C. Noon: 4 °C. By midnight it falls 13 °C from noon. How much did it rise by noon? Midnight temperature?

LO2 I do: subtract a negative

I do

Work out 5 − (−3)

$5 - 2 = 3$

$5 - 1 = 4$

$5 - 0 = 5$

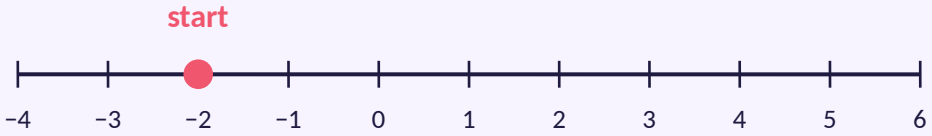
$5 - (-1) = 6$

$5 - (-2) = 7$

LO2 I do: example 2

I do

Work out −2 − (−6)



You say it first, then click.

$4 - (-2)$

$-1 - (-5)$

$-7 - (-3)$

$0 - (-9)$

LO2 Mini whiteboards (1 of 2)

MWB

Write your answer, show me, then click.

$6 - (-4)$

$-8 - (-2)$

LO2 Mini whiteboards (2 of 2)

MWB

Write your answer, show me, then click.

$-3 + (-4)$

Ella writes $-5 - (-2) = -7$. Explain her mistake.

LO2 You do

B · S · G · P

BRONZE

$3 - (-5)$

SILVER

$-6 - (-1)$

GOLD

$-2 - (-7) + (-4)$

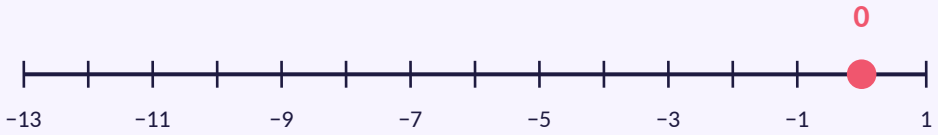
PLATINUM

Find the missing number: $\square - (-4) = 1$

LO3 I do: multiply and divide

I do

Work out −4 × 3



LO3 I do: example 2

I do

Work out −24 ÷ −4

× ÷	+	−
+	+	−
−	−	+

LO3 We do

We do

You say it first, then click.

-5×2

-3×-6

$20 \div -5$

$-18 \div -3$

LO3 Mini whiteboards (1 of 2)

MWB

Write your answer, show me, then click.

-7×4

$-36 \div 9$

LO3 Mini whiteboards (2 of 2)

MWB

Write your answer, show me, then click.

$(-2)^2$

Find two numbers that multiply to −12 and add to 1.

LO3 You do

B · S · G · P

BRONZE

-6×3

SILVER

$-40 \div -8$

GOLD

$(-3) \times (-2) \times (-5)$

PLATINUM

Find the missing number: $\square \times (-6) = 42$

Exam questions

Exam style

1. **Work out −7 + 12**

(1 mark)

2. **Work out −3 × −8**

(1 mark)

3. **The temperature in Moscow is −8 °C. London is 11 °C warmer than Moscow. Oslo is 3 °C colder than Moscow. Find the temperature in London and in Oslo.**

(2 marks)

Exit ticket

5 minutes

On your own. Also on your OnMaths page, where it is marked for you.

1. $-7 + 3$

2. $-2 - (-6)$

3. $-24 \div -4$

Finished early? Extension

Extension

Work out $(-3)^3$

Find the mean of −4, 7, −2 and 3.

Find the missing number: $-2 \times \square - (-5) = 17$

Homework

Due next lesson

5 questions on negative numbers (5 marks).

Do it online on your OnMaths page – it marks itself.
Watch the 2-minute video on your page if you get stuck.
Show your working on paper.

Today I learned: add and subtract ✓ subtract a negative ✓ multiply and divide ✓